QR Water ATM — Detailed Project Report

Oct 14, 2026 · @Ranjeet

1. Executive summary

The QR Water ATM is a self-service chilled-water vending machine: a customer scans a QR code, pays by UPI, and the machine dispenses a measured volume of cold drinking water with no operator and no cash.

It joins two pieces of work that already exist in the project folder:

- **The chiller.** The reference report (NIT Jamshedpur, Mechanical Engineering, Spring 2026) describes a mobile vapour-compression water chilling unit: a 110 W domestic compressor, R-600a refrigerant, a copper-coil evaporator inside a steel dispensing tank, a spiral condenser, and a wheeled 760 × 610 × 460 mm frame with inverter and battery.
- **The payment demo.** A Node/Express backend creates a Razorpay test order of ₹5, verifies the checkout signature, and queues a one-time authorization. An ESP32-WROOM-32 polls the backend and blinks its LED when a payment is verified.

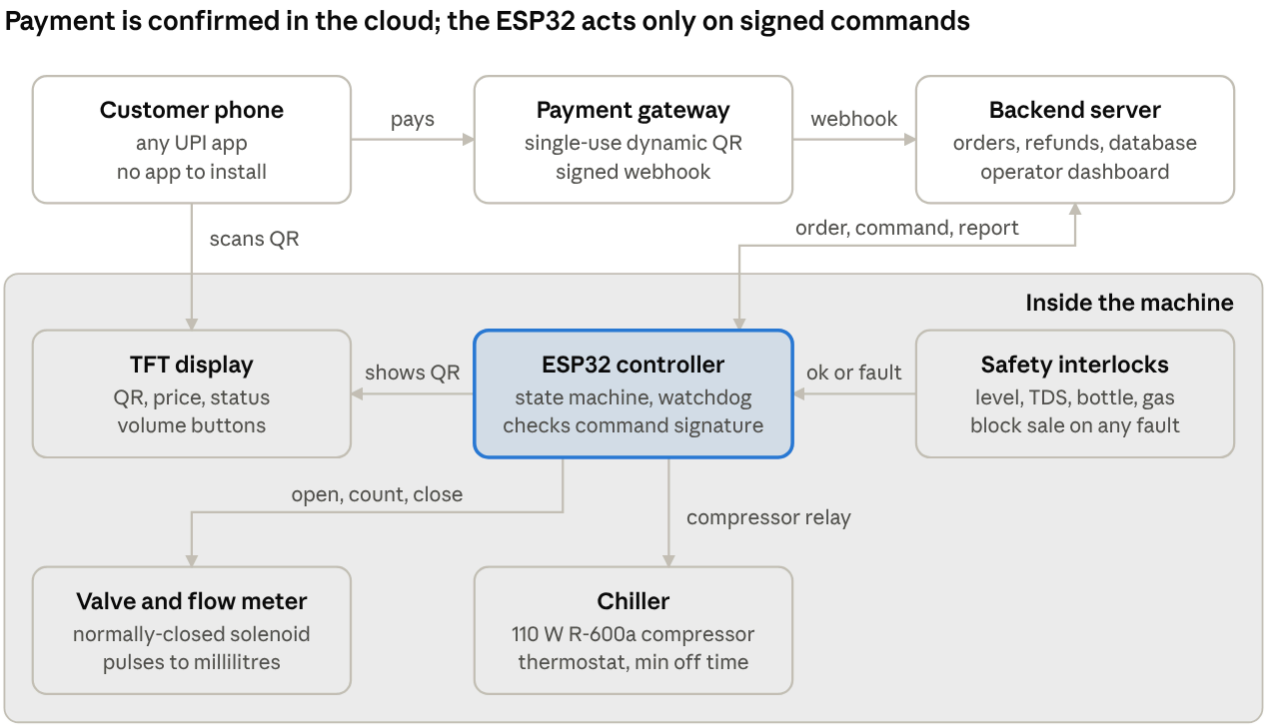
Where the project stands today: payment-to-device signalling works in test mode on a local network. Nothing dispenses water yet. The demo uses a browser checkout, not a QR code; it stores orders in memory; and the ESP32 is not connected to a valve, a flow sensor or the compressor.

What remains, in order: replace the LED with a valve and flow meter, replace browser checkout with a per-transaction dynamic QR confirmed by webhook, add chiller control and safety interlocks, move the backend to a hosted server with a database, then run a field trial.

What the chiller can deliver. The reference data gives a cooling capacity of 0.16 to 0.88 kW, falling as the water gets colder. At a mid value of 0.3 kW, a 20 L tank cools from 30 °C to 10 °C in about 93 minutes, and steady state supports roughly 50 glasses of 250 mL per hour. These are my estimates from the report's ideal-cycle figures; the real machine will do less, so measure it in Phase 1.

2. System architecture

The machine never decides that a payment happened; the backend does, from the gateway's signed webhook, and the ESP32 dispenses only on a signed command.



system architecture · 8 parts, cloud and machine

Read it left to right along the top for money, then down into the machine for water. The customer's phone and the machine never talk to each other directly, which is what makes a fake payment screen useless. Today's demo has the backend and the ESP32 only; the display, valve, flow meter, interlocks and chiller control are still to be added.

3. Hardware and bill of materials

The electronics that turn the chiller into a vending machine cost roughly ₹7,000 to ₹12,000. Prices are approximate Indian retail figures from memory, not quotes; confirm with a local supplier before buying.

Part	Job in the machine	Approx. cost (₹)
ESP32-WROOM-32 dev board (already in use)	Main controller, Wi-Fi, talks to backend	350–500
12 V normally-closed solenoid valve, food grade, 1/2 inch	Opens only while dispensing; closes on power loss	400–700
Hall-effect flow sensor (YF-S201 or food-grade equivalent)	Counts pulses to measure volume delivered	200–350

Part	Job in the machine	Approx. cost (₹)
DS18B20 waterproof probes × 3	Tank water, evaporator, condenser temperatures	300
Tank level sensor (float switch or JSN-SR04T ultrasonic)	Blocks sales when the tank is low	150–350
TDS sensor module	Blocks sales when water quality is out of range	400–600
2.4 to 3.5 inch TFT display	Shows the dynamic QR, price, status	450–900
Relay or contactor rated for compressor start current, with opto-isolation	Switches the 110 W compressor	300–600
MOSFET or relay driver + flyback diode	Drives the solenoid valve	60–100
PZEM-004T or ACS712 current sensor	Measures compressor power for COP and fault detection	250–700
IR or ultrasonic bottle-presence sensor	Dispense only when a bottle is under the nozzle	50–250
Hydrocarbon gas sensor (MQ-6 class)	Detects an R-600a leak and cuts power	150–250
UV-C LED module at the nozzle (optional)	Sanitises the outlet between uses	300–800
4G LTE module (A7670C class) + SIM (optional)	Internet where there is no Wi-Fi	900–1,800
12 V 5 A SMPS + 5 V buck converter	Powers valve, sensors and ESP32	400–600
MCB, fuse, earth-leakage breaker, terminal blocks, IP-rated box	Electrical protection and enclosure	1,000–2,000
Food-grade silicone tubing, SS 304 fittings, push buttons, wiring	Plumbing and assembly	600–1,200

The chiller itself, per the reference report: R-600a compressor (110 W), steel dispensing tank, copper evaporator coil, capillary tube, filter dryer, spiral condenser, base plate (2.25 ft × 1 ft), square-pipe frame, 7.5 cm wheels, inverter and battery.

Open question: do you have access to that fabricated chiller, or does a new one need to be built? The plan in section 5 assumes it exists.

4. Software and payment flow

The production flow replaces browser checkout with a one-time QR shown on the machine, and replaces the browser's confirmation with a server-to-server webhook.

What exists now

File	What it does today
`server.js`	`POST /api/create-order`, `POST /api/verify-payment` (HMAC-SHA256 signature check), `GET /api/device/next`, `POST /api/device/ack`, with a shared bearer token for the device
`public/index.html`	A "Pay ₹5 in Test Mode" button that opens Razorpay Standard Checkout
`firmware/esp32_razorpay_test.ino`	Joins Wi-Fi, polls the backend, blinks GPIO 2 on authorization

Target transaction, step by step

1. The customer presses a volume button (250 mL, 500 mL or 1 L) on the machine.
2. The ESP32 checks its interlocks: tank level, water quality, bottle present, no fault.
3. The ESP32 asks the backend for an order. The backend creates a single-use, fixed-amount UPI QR through the payment gateway, valid for about 2 minutes.
4. The display shows that QR. The customer scans it with any UPI app and pays.
5. The gateway calls the backend webhook. The backend verifies the webhook signature and marks the order paid.
6. The backend sends the machine a dispense command carrying the order id, the volume and a signature.
7. The ESP32 opens the valve, counts flow pulses, and closes the valve at the target volume or on timeout.
8. The ESP32 reports the volume actually delivered. The backend closes the order, or refunds if delivery fell short.

Changes needed in the code

- **Dynamic QR instead of checkout.** Use the gateway's QR API with single-use and fixed-amount options. A printed static QR cannot be tied to one dispense, which the README already warns about.

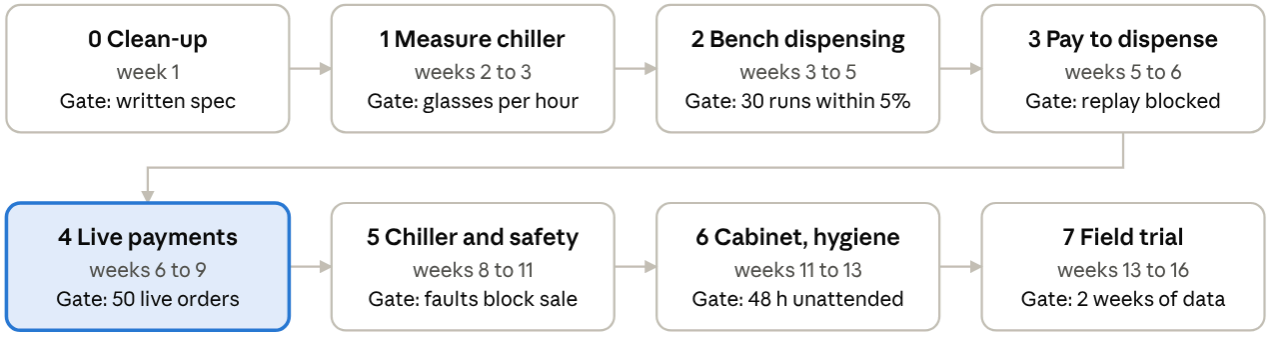
- **Webhook as the source of truth.** Add a webhook endpoint with signature validation. Never trust a success message from the customer's phone.
- **Database.** Replace the in-memory `Map` and array with SQLite or PostgreSQL: tables for machines, orders, dispenses, telemetry and refunds.
- **Idempotency.** One order id can trigger exactly one dispense, even if the webhook arrives twice or the device retries.
- **Push instead of polling.** Move the device link from HTTP polling to MQTT over TLS, so dispensing starts within about a second of payment.
- **Per-device identity.** Give each machine its own key instead of one shared `DEVICE_TOKEN`, and sign commands with a nonce and counter.
- **HTTPS hosting.** Deploy to a public server with TLS; the gateway cannot reach a laptop on a LAN.
- **Firmware state machine.** Idle → Awaiting payment → Dispensing → Reporting → Idle, plus a Fault state, with a hardware watchdog.
- **Operator dashboard.** Sales per day, litres dispensed, tank level, water temperature, compressor energy, alerts.
- **OTA updates.** Signed firmware updates so machines in the field can be fixed remotely.

Security note found while reading the folder: `.env.example` contains what looks like a real Razorpay test key id and secret. Regenerate that key pair in the Razorpay dashboard, keep real values only in `.env`, and put placeholders in the example file before the folder goes into Git or is shared.

5. Step-by-step plan to complete the project

Eight phases take the project from the LED demo to a working machine in about 14 to 16 weeks. Durations are my estimates for a small student team working part time. Each phase ends with a gate: do not start the next phase until the gate is passed.

Sixteen weeks from LED demo to field trial, one gate per phase



roadmap · 8 phases with gates

Phase 4, live payments, is highlighted because it depends on gateway KYC approval, which is outside your control; apply for it during Phase 1.

Phase 0 — Clean-up and decisions (week 1)

- [ ] Regenerate the Razorpay test keys and replace the values in `.env.example` with placeholders
- [ ] Put the folder under Git with `.env` and `node_modules` ignored
- [ ] Decide volumes and prices (for example 250 mL, 500 mL, 1 L)
- [ ] Confirm access to the chiller and check it still runs and holds charge
- [ ] Start a dated lab notebook; it is evidence of invention dates for a patent

Gate: written one-page specification: volumes, prices, target water temperature, site power source.

Phase 1 — Measure the chiller (weeks 2–3)

- [ ] Fit DS18B20 probes on tank water, evaporator and condenser
- [ ] Log temperature every 10 s with the ESP32 while cooling 10 L and 20 L from ambient
- [ ] Measure compressor power with the PZEM or a plug-in meter
- [ ] Compute real cooling capacity and COP from water temperature drop, not from refrigerant tables
- [ ] Record pull-down time and how fast the tank warms when the compressor is off

Gate: a measured figure for glasses per hour at the target temperature. This number sets price and tank size.

Phase 2 — Dispensing on the bench, no payment (weeks 3–5)

- [ ] Wire the solenoid valve through a MOSFET or relay with a flyback diode
- [ ] Wire the flow sensor to an interrupt pin and count pulses
- [ ] Calibrate pulses per litre against a measuring jug at 250 mL, 500 mL and 1 L, ten runs each
- [ ] Add bottle-presence sensor, tank level sensor and an emergency stop button
- [ ] Write the firmware state machine with a dispense timeout and a watchdog

Gate: 30 consecutive dispenses within ±5% of target volume, and the valve closes when power is pulled mid-dispense.

Phase 3 — Join the existing payment demo to dispensing (weeks 5–6)

- [ ] Replace the LED blink in the firmware with the dispense routine

- [ ] Send the purchased volume in the authorization from `server.js`
- [ ] Report delivered volume in the `ack` call
- [ ] Test end to end in Razorpay test mode on the local network

Gate: a test payment produces a measured dispense, and replaying the same order id dispenses nothing.

Phase 4 — Production payment path (weeks 6–9)

- [ ] Complete gateway KYC for a live account (needs a bank account and business or individual proof)
- [ ] Switch to single-use dynamic QR and render it on the TFT display
- [ ] Add the webhook endpoint with signature validation
- [ ] Move storage to a database and add idempotency keys
- [ ] Deploy the backend on a hosted server with HTTPS
- [ ] Move device communication to MQTT over TLS with a per-device key
- [ ] Add automatic refund when delivered volume is short

Gate: 50 live ₹1 transactions with zero unpaid dispenses and zero paid-but-not-dispensed orders left unrefunded.

Phase 5 — Chiller control and safety (weeks 8–11)

- [ ] Thermostat control of the compressor with hysteresis (for example on at 12 °C, off at 8 °C)
- [ ] Minimum 3 to 5 minutes off time before any compressor restart
- [ ] Low-temperature cut-out so the coil cannot freeze the tank
- [ ] Gas sensor near the refrigerant circuit that cuts compressor power on a leak
- [ ] Separate the 230 V section from low-voltage electronics; add MCB, earth-leakage breaker and proper earthing of the frame
- [ ] Water-quality interlock on TDS; hide the QR when quality or level is out of range

Gate: each fault is injected by hand and the machine refuses to sell and raises an alert.

Phase 6 — Enclosure, hygiene and dashboard (weeks 11–13)

- [ ] Mount display, buttons and nozzle on a front panel; add a drip tray and drain
- [ ] Insulate the tank and cold lines to stop condensation and heat gain
- [ ] Use food-grade tubing and fittings on every wetted part
- [ ] Build the operator dashboard and SMS or WhatsApp alerts
- [ ] Write the cleaning procedure and schedule

Gate: the machine runs unattended for 48 hours in the lab.

Phase 7 — Field trial and report (weeks 13–16)

- [ ] Place the machine at one site, such as a hostel or canteen, for two weeks
- [ ] Log sales, uptime, failures, energy per litre and customer feedback
- [ ] Send a water sample to a lab for testing against the drinking-water standard
- [ ] File the provisional patent before any public demonstration or report submission (section 8)
- [ ] Write the final report in the format of the reference PDF

Gate: two weeks of field data and a list of fixes for version 2.

6. Challenges and solutions

The three that can sink the project are payment-to-dispense matching, R-600a safety next to switching electronics, and cooling capacity at peak demand. The rest are routine engineering.

Area	Challenge	Solution
Payment	A static printed QR cannot tell which payment belongs to which dispense	Use a single-use dynamic QR per order with a fixed amount and a 2-minute expiry
Payment	Customer pays but the machine is offline, so no water comes out	Hold the order as paid-pending for 60 s, retry, then refund automatically; show a help number on the display
Payment	Duplicate webhooks or device retries cause double dispensing	Idempotency: the order row moves created → paid → dispensing → done exactly once; the device stores the last order ids in flash
Payment	Fake “payment success” screens or replayed commands	Trust only the signed webhook; sign device commands with a per-device key, nonce and counter
Payment	Gateway fee and KYC on ₹5 tickets	Check the current fee schedule before pricing; offer a prepaid wallet or larger volumes to cut per-transaction cost

Area	Challenge	Solution
Connectivity	Weak Wi-Fi or mobile signal at the site	4G module with external antenna, automatic reconnect, and the offline token idea in section 7
Dispensing	Flow sensor drifts with pressure and tank level	Calibrate at several tank levels; use a small pump for constant pressure, or a load cell under the bottle as a second check
Dispensing	Valve sticks open or the flow sensor fails	Normally-closed valve, hard timeout in firmware, and a second valve in series for a commercial unit
Dispensing	Power cut in the middle of a dispense	Save progress to flash every 50 mL; on restart report the partial volume and refund the balance
Dispensing	No bottle, or the bottle overflows	Bottle-presence sensor, stop button, and an overflow tray with a level switch
Chiller	Demand at peak hours exceeds the 110 W compressor's capacity	Size the tank from the Phase 1 measurements; pre-chill before peaks; add thermal storage (section 7)
Chiller	Compressor short-cycling damages it	Enforce 3 to 5 minutes minimum off time and a wide thermostat band
Chiller	Evaporator runs below 0 °C (the reference data reaches −6 °C) and ice forms on the coil	Low-temperature cut-out on the coil probe; gentle stirring or recirculation in the tank
Chiller	High ambient temperature raises condenser temperature and COP falls (reference data: COP 8 at 32 °C down to 1.4 at 56 °C)	Add a condenser fan, shade and clear airflow; keep the condenser clean
Safety	R-600a is flammable and relays can spark	Sealed relay or solid-state switching, electronics in a separate ventilated compartment, gas sensor cut-off, brazed joints leak-tested
Safety	230 V and water in the same cabinet	Earth-leakage breaker, earthed frame, IP-rated electrical box above the wet zone, drip loops on cables

Area	Challenge	Solution
Electronics	Compressor start causes noise that resets the ESP32	Opto-isolated relay, separate supply for logic, snubber across the contacts, brown-out detection and watchdog
Hygiene	Biofilm in the tank and at the nozzle	Food-grade stainless tank, scheduled cleaning, UV-C at the outlet, a recessed nozzle that bottles cannot touch
Water quality	Feed water quality varies by site	Pre-filter and RO or UV stage as the site requires; TDS interlock; periodic lab test
Operations	Vandalism, theft and tampering	Locked steel cabinet, door switch alert, no cash inside the machine
Regulatory	Selling drinking water and dispensing by volume carry obligations	Ask the institute or a local consultant about food-safety registration, the drinking-water standard and legal-metrology rules before commercial sale

7. Innovation ideas that could lead to a patent

A QR-paid water ATM on its own is not new: UPI water vending machines are already sold in India, so that combination will not be granted. A patent has to rest on a specific technical feature that prior machines lack. The strongest candidates come from joining the payment data to the refrigeration cycle, because that is where this project has both halves and most vendors do not.

I have not run a prior-art search, so every novelty remark below is my judgment, not a finding. Search before investing in any of them (section 8).

Recommended three to pursue

A. Demand-predictive chilling with thermal storage. The controller learns the sales pattern from payment timestamps and pre-chills before predicted peaks. A phase-change material or ice bank around the evaporator stores cold while solar power or cheap power is available, so a 110 W compressor can serve a peak several times its steady capacity.

- Technical effect: smaller compressor, lower energy per litre, fewer starts.

- Build: PCM packs or an ice-bank jacket on the tank, an extra temperature probe, a forecasting routine on the backend, a schedule pushed to the ESP32.
- Claim direction: an apparatus in which compressor operation is scheduled from transaction history and state of charge of a thermal store.

B. Verified-delivery closed loop with automatic partial refund. Two independent measurements (flow pulses and bottle weight from a load cell) must agree before a transaction closes. A shortfall triggers a proportional refund without human action, and persistent disagreement flags a sensor fault.

- Technical effect: metering accuracy and self-diagnosis without a certified meter.
- Build: load cell with HX711 under the bottle platform, comparison logic in firmware, refund call on the backend.
- Claim direction: a dispensing system that reconciles two volume measurements and settles payment against the delivered quantity.

C. Offline-tolerant signed dispense token. The machine shows a QR containing its id and a fresh nonce. After payment the server issues a short signed token to the customer's phone, which passes it to the machine over Bluetooth or as a QR read by a small camera. The machine verifies the signature and nonce locally, so it dispenses even when its own internet link is down.

- Technical effect: reliable vending in poor-coverage sites, with replay protection.
- Build: Ed25519 verification on the ESP32, BLE or an ESP32-CAM reader, a monotonic counter in flash.
- Claim direction: a vending controller that authorizes dispensing from a customer-carried cryptographic token bound to a machine-generated nonce.

Further ideas

Idea	What would be new	Main parts	My view of patent prospects
Temperature-selectable dispensing billed by cooling delivered	A mixing valve blends chilled and ambient water to the chosen temperature; the charge tracks heat removed	Proportional mixing valve, outlet probe	Fair if claimed as the mixing-control apparatus; pricing alone is a business method and excluded

Idea	What would be new	Main parts	My view of patent prospects
Condenser waste-heat recovery for line sanitisation	Heat rejected by the condenser warms a small reservoir used to hot-flush the dispensing line on a schedule	Plate heat exchanger, flush valve	Fair; heat recovery is known, tying it to automatic hygiene flushing may be distinct
Sale lock on water quality	The QR is not generated when TDS, turbidity or filter life is out of range, so payment cannot be taken for unsafe water	TDS and turbidity sensors	Weak to fair; simple interlocks are often judged obvious
Live COP self-diagnosis	The controller estimates COP from water temperature slope and compressor power, and flags refrigerant loss or condenser fouling	Power sensor, three probes	Fair; fault detection in refrigeration is a crowded field
R-600a leak-safe vending cabinet	Gas sensing with a staged response: cut compressor, run a vent fan, lock sales, alert operator	Gas sensor, fan, sealed switching	Fair as a safety arrangement for hydrocarbon vending units
Contactless nozzle with UV-C cycle keyed to each sale	UV exposure runs between transactions and is logged against the order	UV-C LED, shield, interlock	Weak; UV nozzles exist in dispensers
Reusable-bottle credit	A tag on the bottle earns a discount and tracks plastic avoided	NFC reader, tags	Weak for a patent; good for a sustainability story

How to raise the odds

- Combine ideas into one system claim. A plus B together, on a hydrocarbon chiller, is harder to find in prior art than either alone.
- Build and measure. A patent needs a working description, and numbers such as energy per litre with and without thermal storage make the technical effect concrete.
- Record everything with dates in the lab notebook: sketches, test data, photos.

8. Patent filing route in India

File a provisional application before the idea is shown to anyone outside the team. Public disclosure before filing, including a project report, a poster, a demo or a social-media post, can destroy novelty.

1. **Keep it confidential.** Tell your guide and the institute's IPR cell only; ask who will own the invention under institute policy.
2. **Search prior art.** Use Google Patents, the Indian Patent Office search (InPASS), Espacenet and Google Scholar. Search terms: water vending, UPI or QR payment, thermal storage chiller, dispense verification, offline token.
3. **Write an invention disclosure.** Problem, existing solutions, what is different, drawings, test results, list of inventors.
4. **File a provisional specification.** It secures the priority date and gives 12 months to finish the work.
5. **File the complete specification within 12 months,** with claims, drawings and an abstract.
6. **Publication and examination.** The application is published after 18 months unless early publication is requested; examination starts only after a request is filed.
7. **Respond to the examination report,** attend a hearing if called, and the patent is granted if objections are cleared.

Points specific to this project

- Indian law excludes computer programs as such and business methods. Write claims around the apparatus: sensors, valves, thermal store, controller and what they do physically. Do not claim "a pricing method" or "an app".
- The chiller design in the reference report belongs to its authors. Your claims must cover what you add, and those authors should be credited or included as inventors if their contribution is part of the claimed invention.
- The cabinet's appearance can be protected separately by a design registration.

Approximate official fees for an individual, startup or educational institution filing online, from memory; check the current fee schedule on the Indian Patent Office website before paying:

Step	Approx. fee (₹)
Filing the application (provisional or complete)	1,600
Request for early publication	2,500
Request for examination	4,000
Request for expedited examination	8,000

A patent agent's drafting charge is extra, commonly tens of thousands of rupees. Many institutes pay filing costs through their IPR cell, so ask there first. I could not look these figures up in this session, so treat them as a starting point.

9. Cost, business model and next actions

Energy is a small cost; the gateway fee, water treatment and maintenance decide whether a ₹5 glass pays.

Energy per glass. Cooling 250 mL by 20 °C removes about 21 kJ of heat. At a real COP of 2, that is roughly 0.003 kWh of electricity, which costs about 2 to 3 paise at ₹8 per kWh. Standby losses through the tank walls will be larger than this, so insulation matters more than compressor efficiency.

Prototype cost. About ₹7,000 to ₹12,000 for the electronics in section 3, plus the chiller if it has to be rebuilt.

Ways to earn

- Pay per volume at the machine: the base case.
- Prepaid wallet or monthly pass for hostels and offices, which cuts per-transaction fees.
- Machine sale or lease to canteens, with a dashboard subscription.
- Sponsorship or advertising on the display and cabinet.

Numbers to measure before quoting a payback period: litres sold per day at the trial site, gateway fee per transaction, filter and cleaning cost per month, and uptime. I have not estimated payback because none of these is known yet.

Next actions this week

- [ ] Regenerate the Razorpay test keys and clean `.env.example`
- [ ] Confirm access to the chiller and its condition
- [ ] Buy the Phase 2 parts: solenoid valve, flow sensor, three DS18B20 probes, relay, 12 V supply
- [ ] Start the dated lab notebook
- [ ] Meet the guide and the IPR cell about ideas A, B and C before any public presentation